

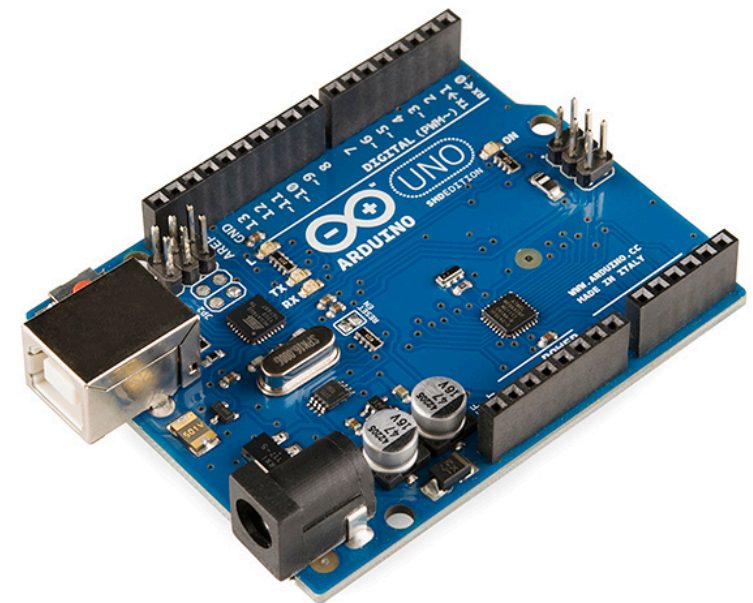
EG1311: Design & Make

Arduino Basics

Dr. Jason S. Ku

Arduino: What is it?

- Open-source platform (large community)
- Hardware
 - Micro-processor computer (ATmega328P)
 - A clock (resonator)
 - A power supply
 - A USB interface
- Software
 - Arduino IDE (editor, compiler, programmer)
 - Arduino Library



Arduino: What is it?

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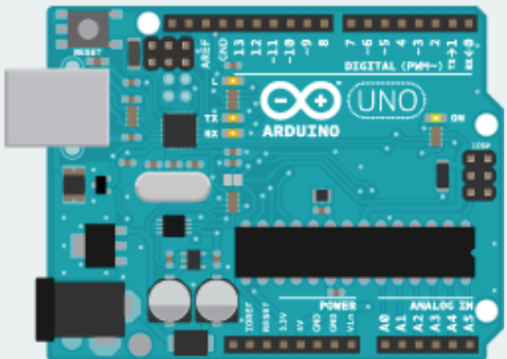
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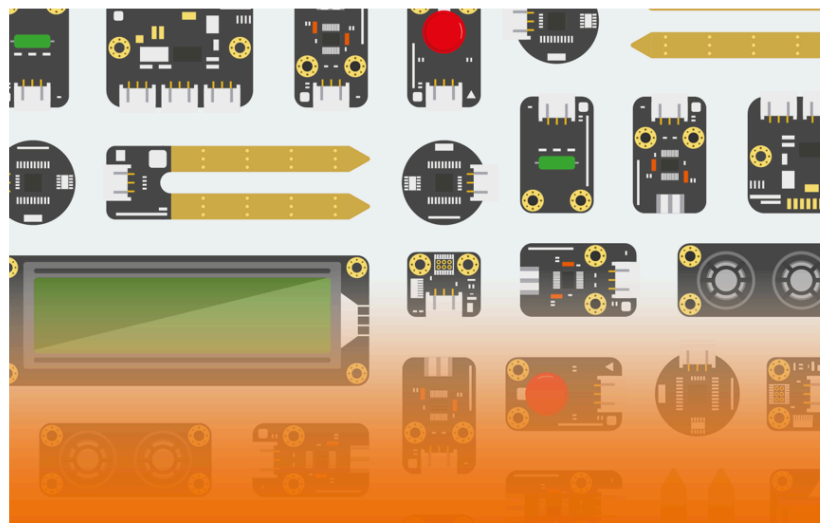


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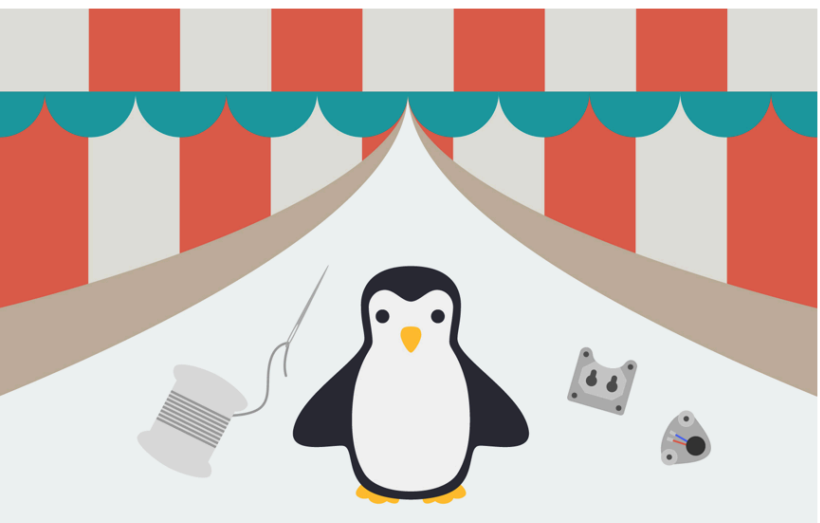
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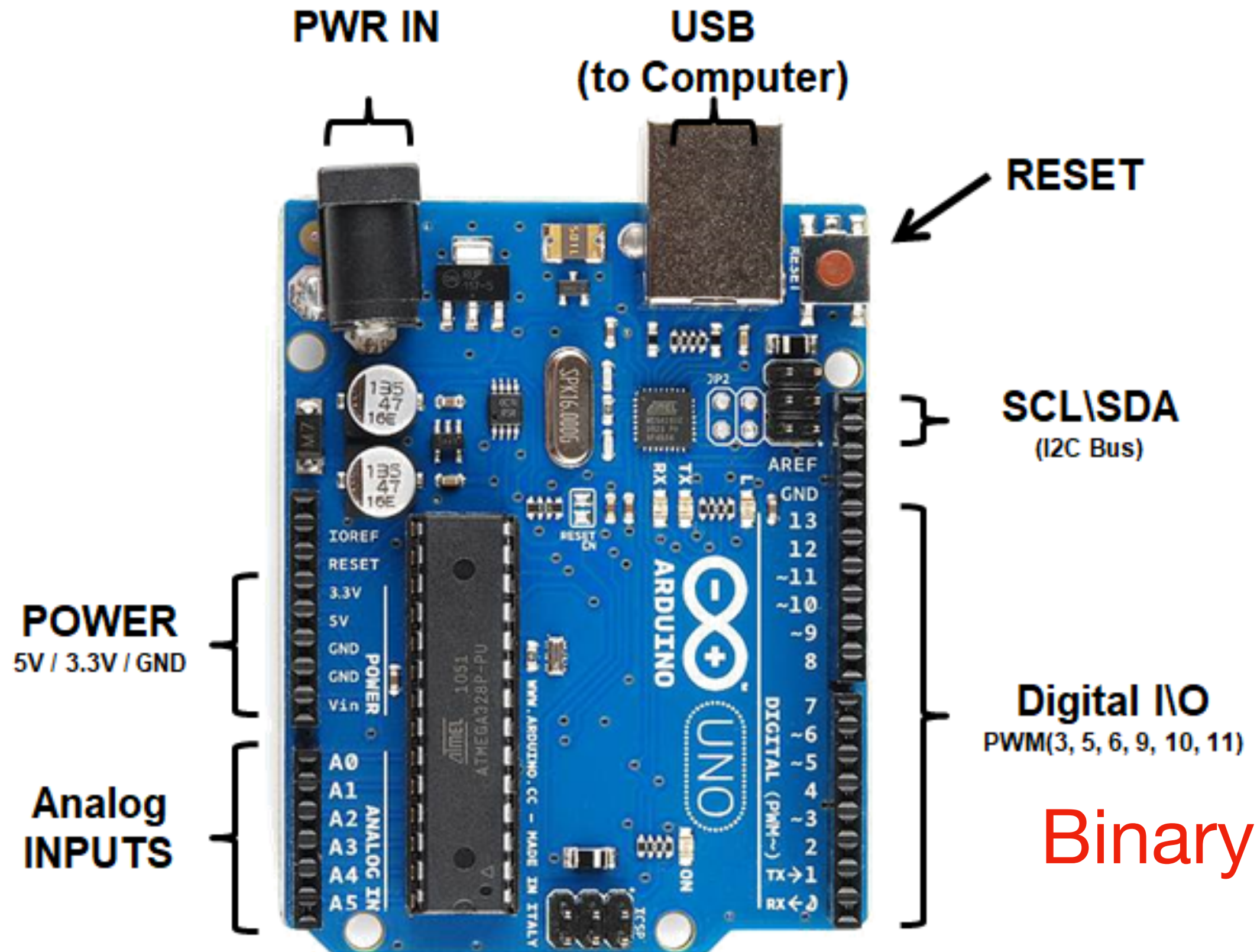
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BLOG

BLOG

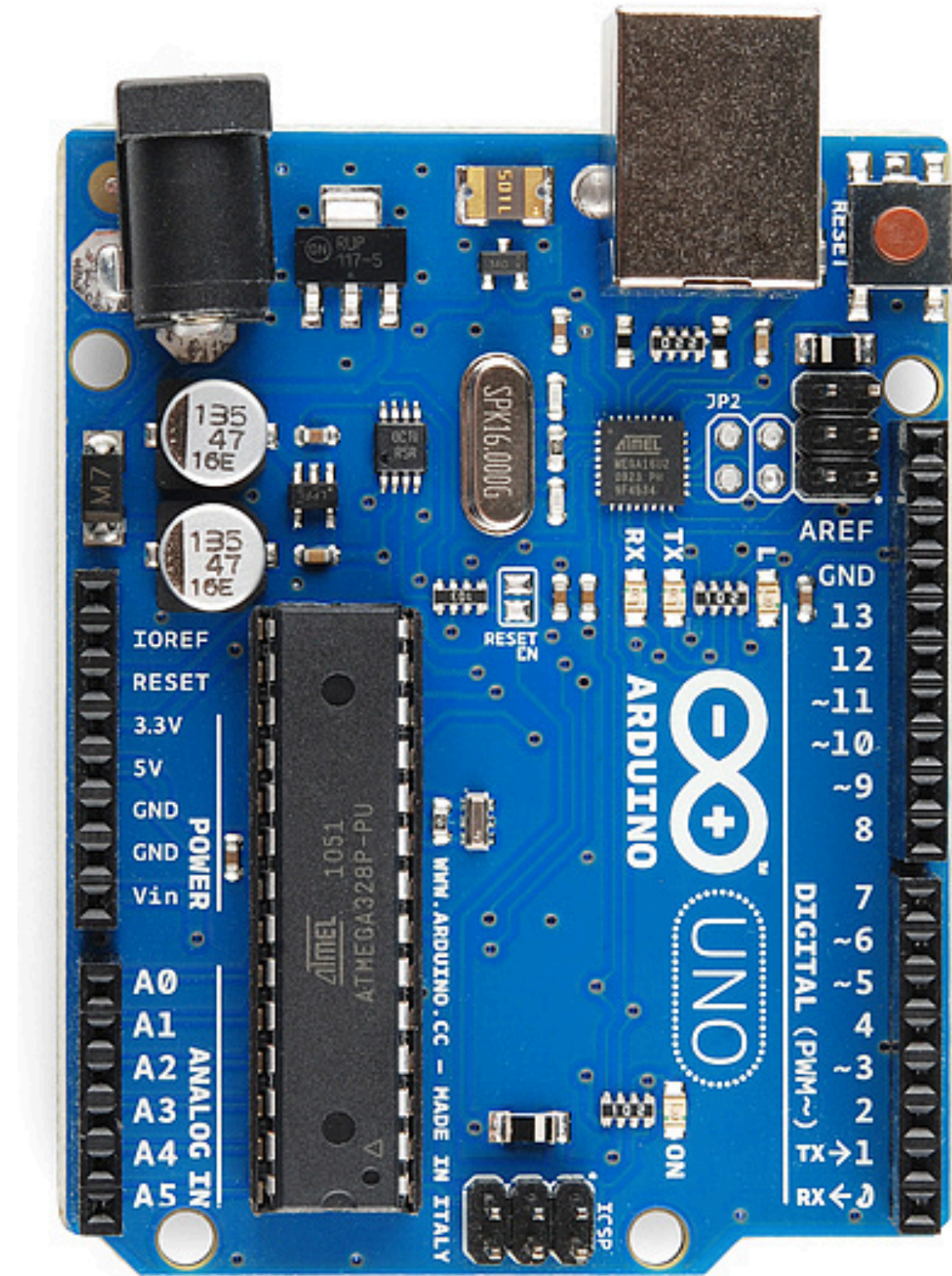
<https://www.arduino.cc/en/software>

Arduino: Hardware Anatomy



Arduino: Hardware Anatomy

- VIN: 7-12 V, ~1A
- USB port: 5V, ~1A
- Digital pins: 5V, ~40mA
- 5V pin: ~800mA
- Pins 0-1: Serial COM
- Pin 13: LED pin
- Pin (~): PWM



Arduino: API

Functions

For controlling the Arduino board and performing computations.

Digital I/O digitalRead() digitalWrite() pinMode()	Math abs() constrain() map() max() min() pow() sq() sqrt()	Random Numbers random() randomSeed() Bits and Bytes bit() bitClear() bitRead() bitSet() bitWrite() highByte() lowByte()
Analog I/O analogRead() analogReference() analogWrite()	Trigonometry cos() sin() tan()	External Interrupts attachInterrupt() detachInterrupt()
Zero, Due & MKR Family analogReadResolution() analogWriteResolution()	Characters isAlpha() isAlphaNumeric() isAscii() isControl() isDigit() isGraph() isHexadecimalDigit() isLowerCase() isPrintable() isPunct() isSpace() isUpperCase() isWhitespace()	Interrupts interrupts() noInterrupts() Communication Serial Stream
Advanced I/O noTone() pulseIn() pulseInLong() shiftIn() shiftOut() tone()		USB Keyboard Mouse
Time delay() delayMicroseconds() micros() millis()		

Variables

Arduino data types and constants.

Constants HIGH LOW INPUT OUTPUT INPUT_PULLUP LED_BUILTIN true false Floating Point Constants Integer Constants	Data Types array bool boolean byte char double float int long short size_t string String() unsigned char unsigned int unsigned long void word	Utilities PROGMEM sizeof() Variable Scope & Qualifiers const scope static volatile
Conversion (unsigned int) (unsigned long) byte() char() float() int() long() word()		

Structure

The elements of Arduino (C++) code.

Sketch loop() setup()	Arithmetic Operators % (remainder) * (multiplication) + (addition) - (subtraction) / (division) = (assignment operator) Comparison Operators != (not equal to) < (less than) <= (less than or equal to) == (equal to) > (greater than) >= (greater than or equal to) Boolean Operators ! (logical not) && (logical and) (logical or)	Pointer Access Operators & (reference operator) * (dereference operator) Bitwise Operators & (bitwise and) << (bitshift left) >> (bitshift right) ^ (bitwise xor) (bitwise or) ~ (bitwise not) Compound Operators %= (compound remainder) &= (compound bitwise and) *= (compound multiplication) ++ (increment) += (compound addition) -- (decrement) -= (compound subtraction) /= (compound division) ^= (compound bitwise xor) = (compound bitwise or)
Control Structure break continue do...while else for goto if return switch...case while Further Syntax #define (define) #include (include) /* */ (block comment) // (single line comment) ; (semicolon) { } (curly braces)		

Arduino: API

- Required Arduino function definitions

```
void setup() {  
    // do at start  
}  
  
void loop() {  
    // repeat forever  
}
```

Arduino: API

- Defined integer constants

true, false // Booleans

HIGH, LOW // Binary Voltage

INPUT, OUTPUT // Pin Mode

Arduino: API

- Basic Digital Input/Output (I/O)

```
void pinMode(pin, OUTPUT);
```

```
void digitalWrite(pin, HIGH);
```

```
int digitalRead(pin);
```

Arduino: API

- Timing

```
void delay(millisecs);
```

```
void delayMicroseconds(microsecs);
```

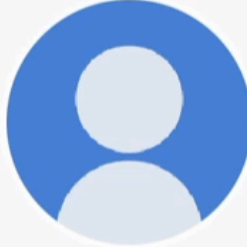
```
unsigned long millis(); // 50 days
```

```
unsigned long micros(); // 1 hour
```

Blink Demo

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eg1311_demo

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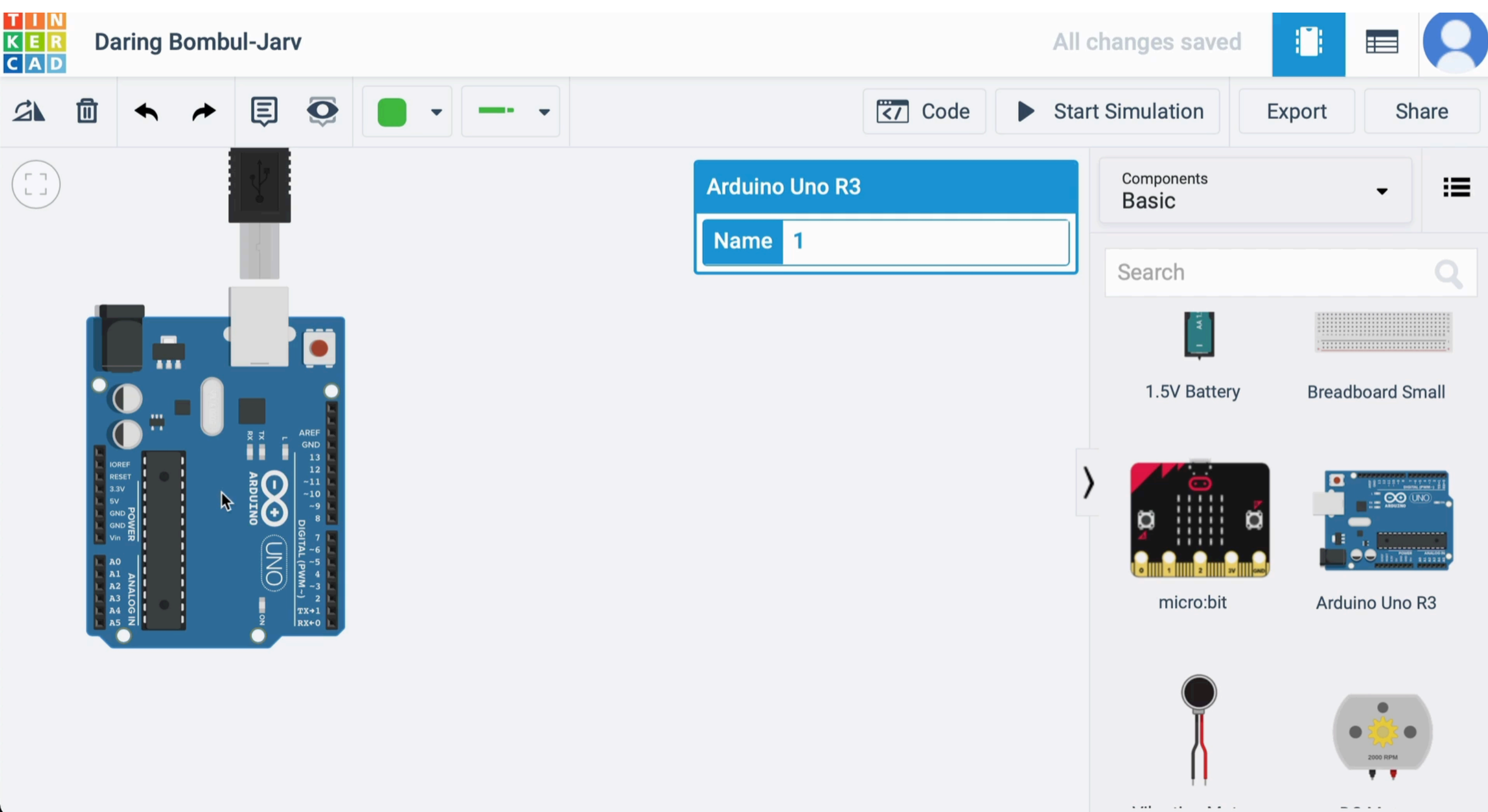
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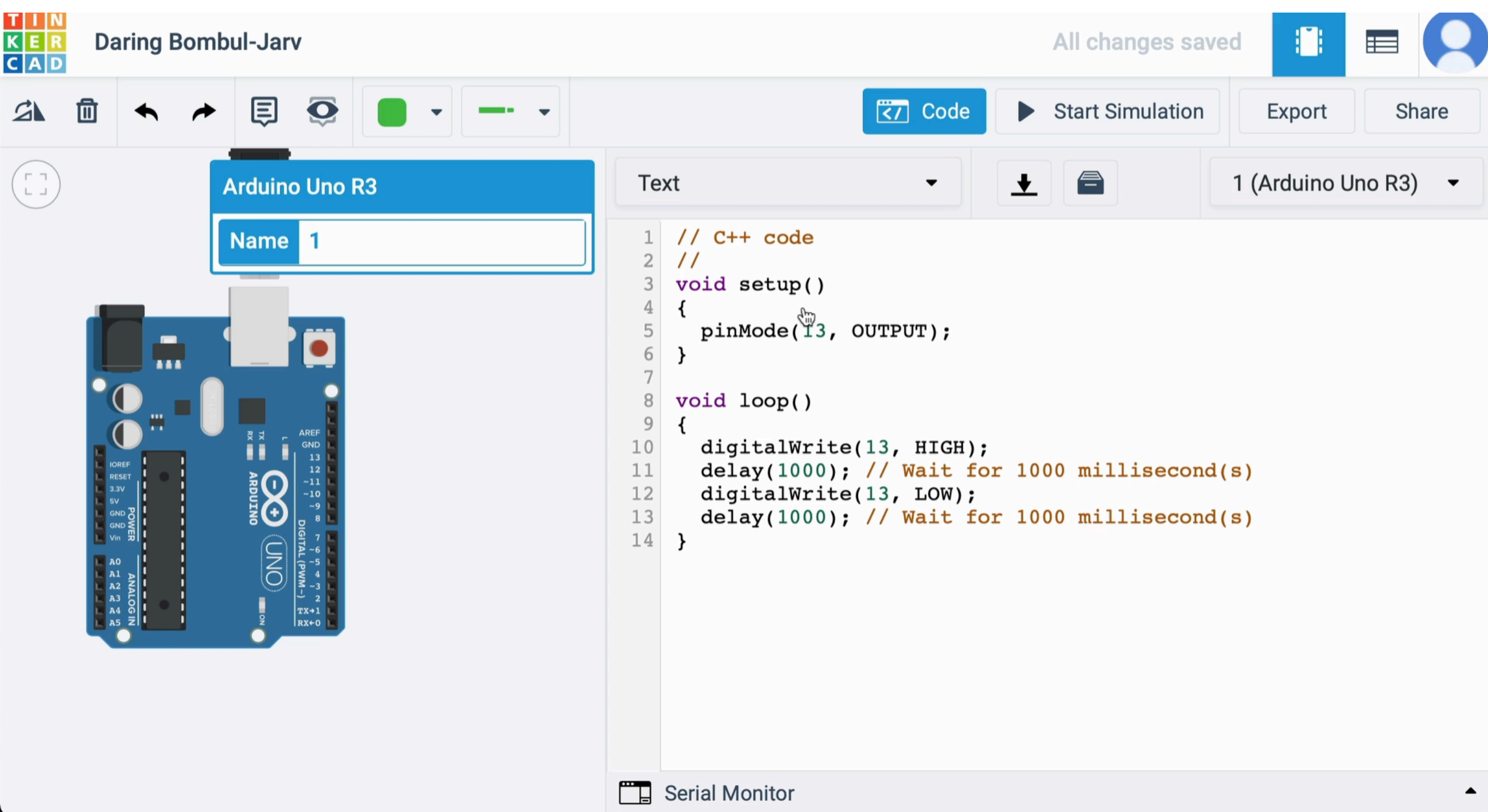


<https://www.tinkercad.com>

Blink Demo

The screenshot displays the Tinkercad web interface. At the top left, the Tinkercad logo is visible, followed by the username "Daring Bombul-Jarv". On the top right, a status bar indicates "All changes saved" and includes icons for a document, a list, and a user profile. Below the username, a toolbar contains icons for undo, redo, delete, and a green button for starting a simulation. The main workspace features a 3D model of an Arduino Uno R3 board on the left. A blue box labeled "Arduino Uno R3" is positioned above the board, with a text input field containing the name "1". To the right of the workspace is a "Blocks" panel with a dropdown menu set to "Blocks". It lists "Blocks + Text" and "Text" (which is selected). Below these are several blue blocks: "set built-in LED to HIGH", "set pin 0 to HIGH", "set pin 3 to 0", "rotate servo on pin 0 to 0 degrees", "play speaker on pin 0 with tone 60", and "turn off speaker on pin 0". On the far right, a "Code" panel shows a sequence of blocks: "set built-in LED to HIGH", "wait 1 secs", "set built-in LED to LOW", and "wait 1 secs". At the bottom of the interface, there is a "Serial Monitor" tab.

Blink Demo

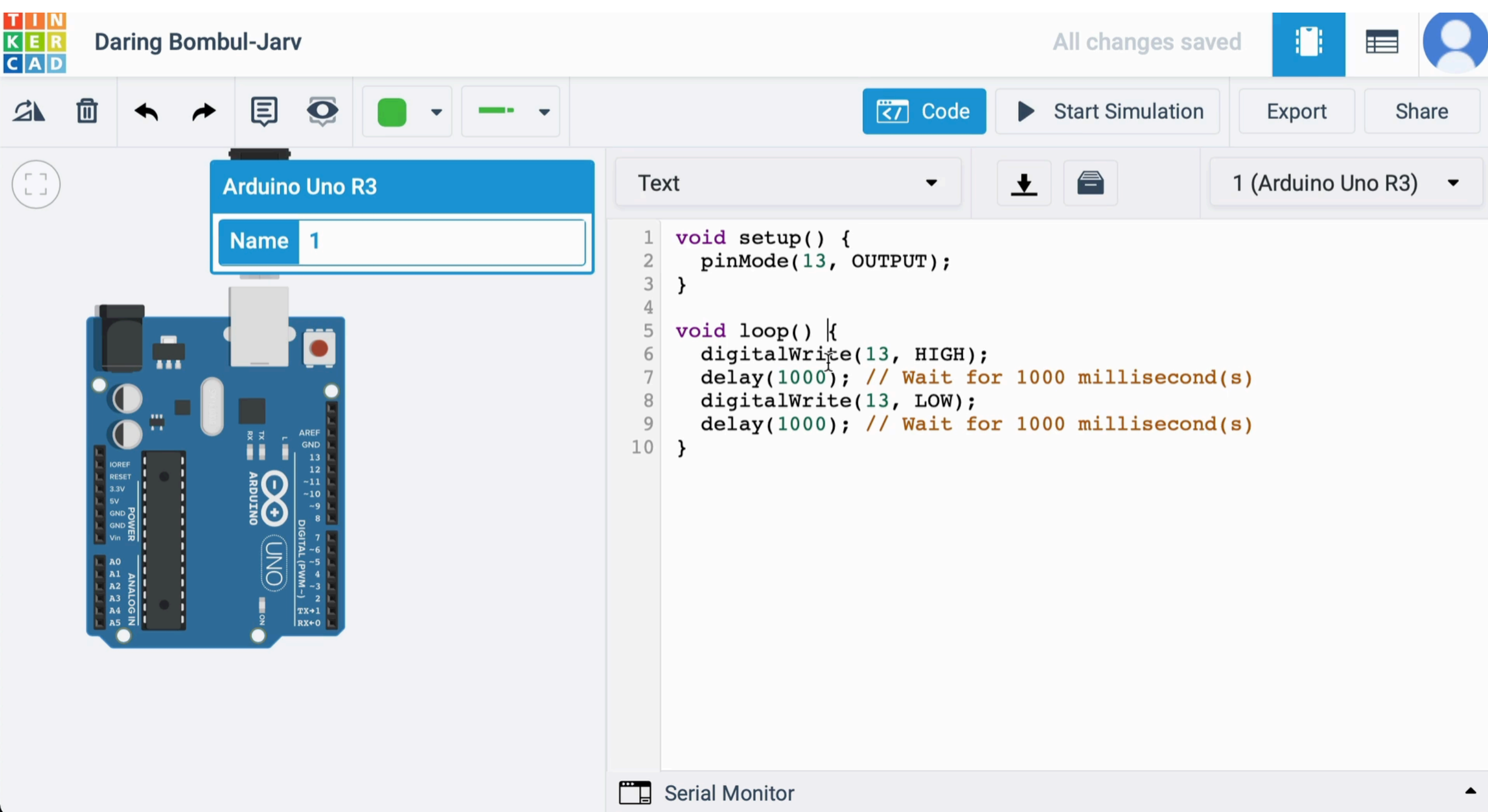


The screenshot displays the Tinkercad web interface. At the top left, the Tinkercad logo is visible, followed by the username "Daring Bombul-Jarv". On the right, a status bar indicates "All changes saved" and includes icons for a document, a list, and a user profile. Below the username, a toolbar contains icons for undo, redo, delete, and a green cube. The main workspace on the left shows an "Arduino Uno R3" board with a "Name" field set to "1". The board is a blue PCB with various components like a USB Type-C port, a USB Type-A port, a DC power jack, and several pins labeled "DIGITAL (PWM ~)" and "ANALOG IN". On the right, a "Code" editor is open, showing C++ code for a blink demo. The code is as follows:

```
1 // C++ code
2 //
3 void setup()
4 {
5     pinMode(13, OUTPUT);
6 }
7
8 void loop()
9 {
10    digitalWrite(13, HIGH);
11    delay(1000); // Wait for 1000 millisecond(s)
12    digitalWrite(13, LOW);
13    delay(1000); // Wait for 1000 millisecond(s)
14 }
```

At the bottom of the interface, a "Serial Monitor" tab is visible.

Blink Demo

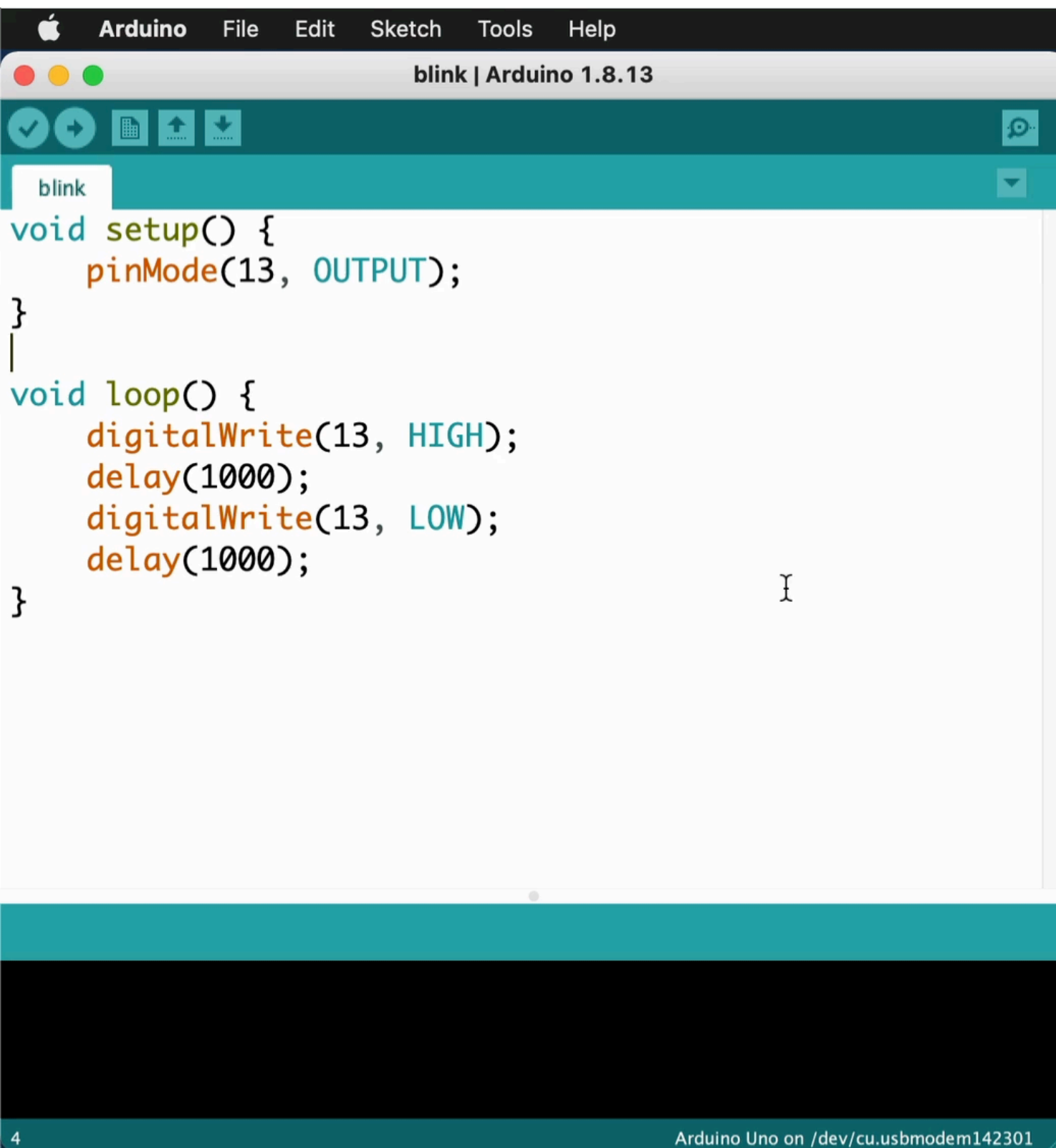


The screenshot displays the Tinkercad web interface. At the top left, the Tinkercad logo is visible, followed by the username "Daring Bombul-Jarv". On the right, a status bar indicates "All changes saved" and includes icons for a document, a list, and a user profile. Below the username, a toolbar contains icons for undo, redo, delete, and a green cube. The main workspace features a 3D model of an Arduino Uno R3. A blue box labeled "Arduino Uno R3" is positioned above the board, with a "Name" field containing the number "1". To the right of the workspace, a "Code" button is present, followed by "Start Simulation", "Export", and "Share" buttons. Below these, a "Text" dropdown menu is shown, and a "1 (Arduino Uno R3)" dropdown menu is visible. The code editor displays the following C++ code:

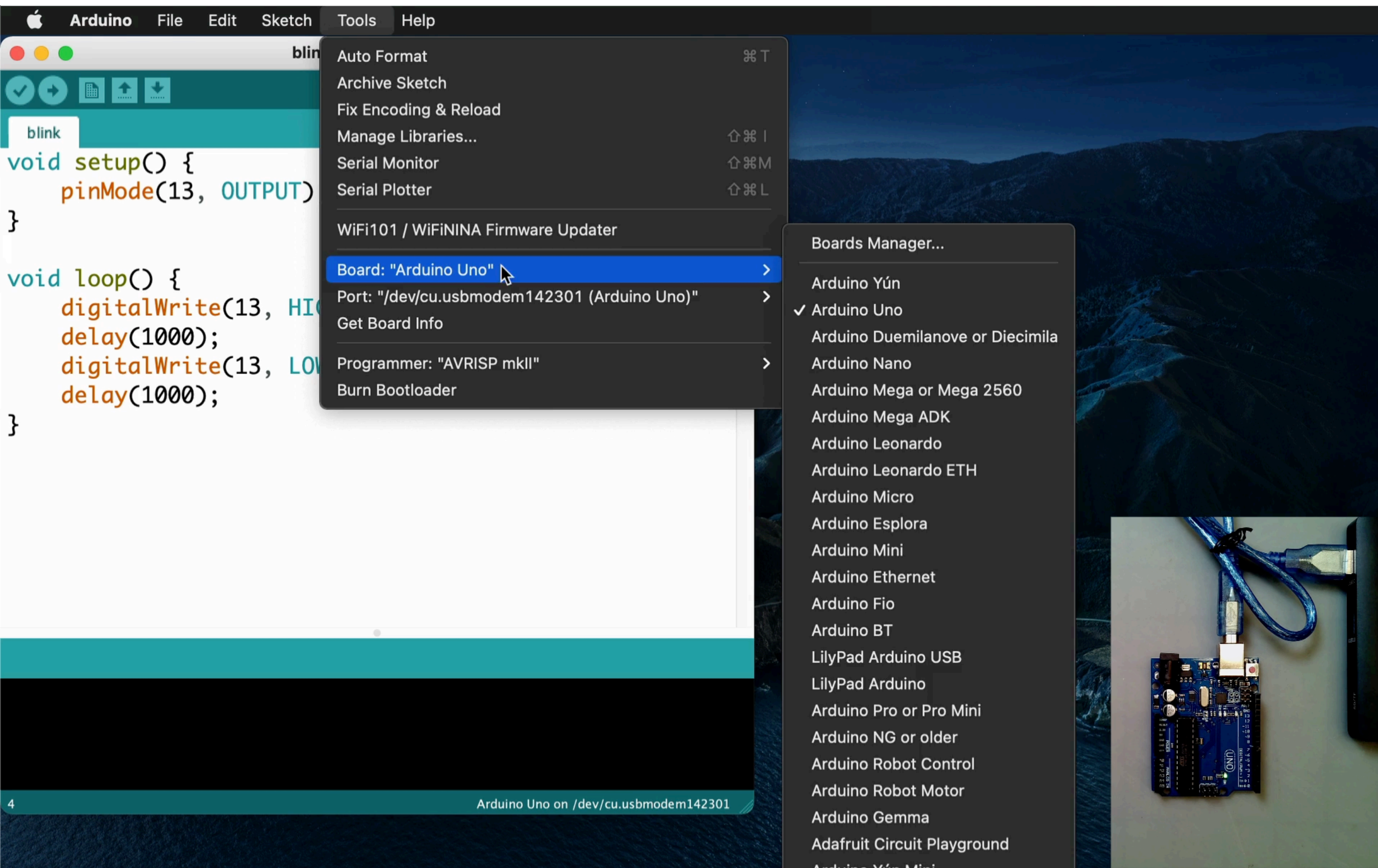
```
1 void setup() {  
2   pinMode(13, OUTPUT);  
3 }  
4  
5 void loop() {  
6   digitalWrite(13, HIGH);  
7   delay(1000); // Wait for 1000 millisecond(s)  
8   digitalWrite(13, LOW);  
9   delay(1000); // Wait for 1000 millisecond(s)  
10 }
```

At the bottom of the interface, a "Serial Monitor" button is visible.

Blink Demo



Blink Demo



The screenshot shows the Arduino IDE interface. The main window displays the 'Blink' sketch, which is a simple program that turns an LED on and off. The code is as follows:

```
void setup() {  
  pinMode(13, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(13, HIGH);  
  delay(1000);  
  digitalWrite(13, LOW);  
  delay(1000);  
}
```

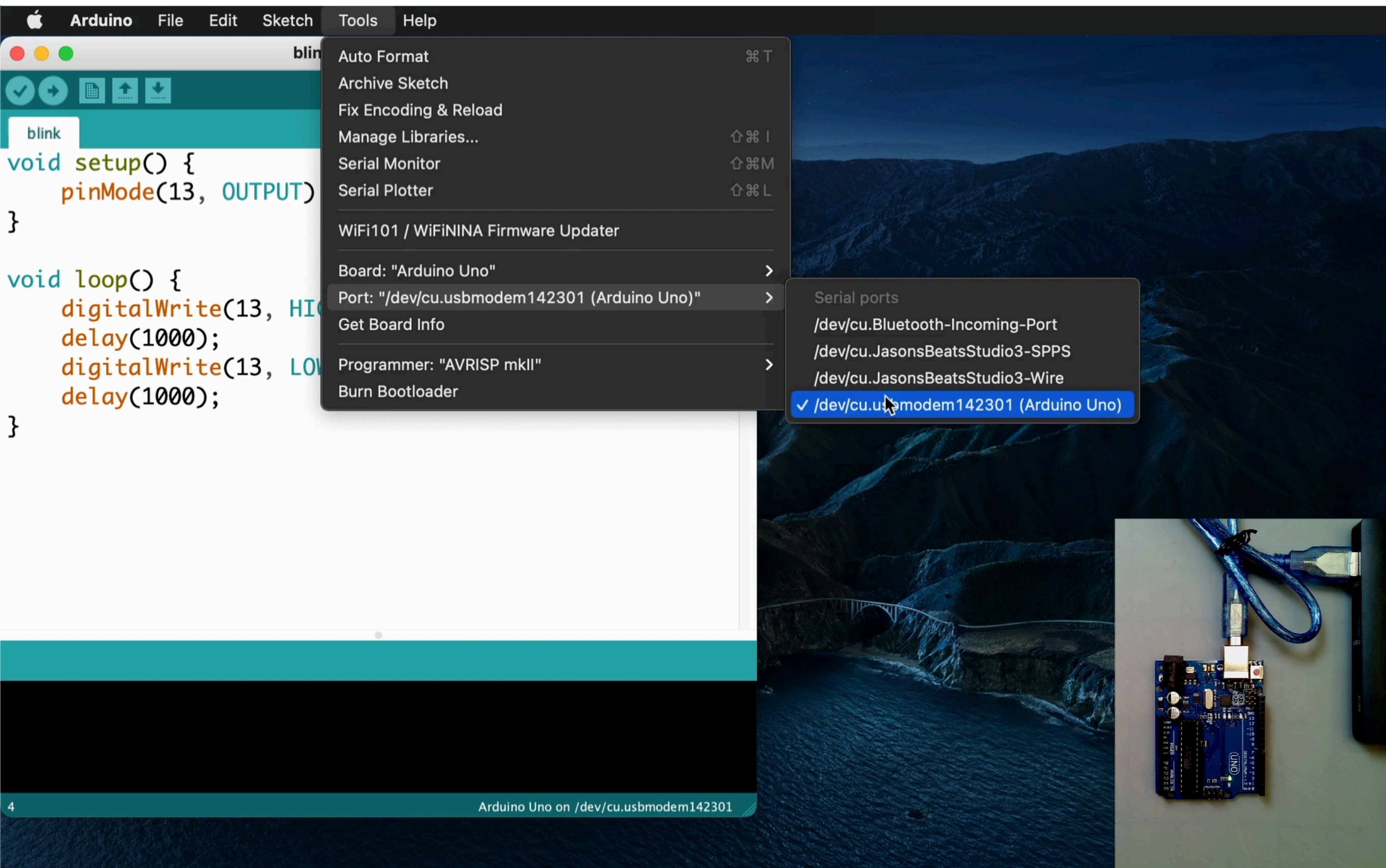
The 'Tools' menu is open, showing various options. The 'Board' dropdown is set to 'Arduino Uno'. The 'Port' dropdown is set to '/dev/cu.usbmodem142301 (Arduino Uno)'. The 'Programmer' dropdown is set to 'AVRISP mkII'. The 'Burn Bootloader' option is also visible.

The 'Boards Manager...' window is open, showing a list of available boards. The 'Arduino Uno' board is selected, indicated by a checkmark.

The status bar at the bottom of the IDE shows '4' on the left and 'Arduino Uno on /dev/cu.usbmodem142301' on the right.

In the bottom right corner, there is a photograph of an Arduino Uno board connected to a USB cable.

Blink Demo



The screenshot shows the Arduino IDE interface on a Mac. The 'Tools' menu is open, displaying various options. The 'Port' option is selected, showing a list of serial ports. The port '/dev/cu.usbmodem142301 (Arduino Uno)' is highlighted with a checkmark. The code editor on the left shows the 'blink' sketch with the following code:

```
void setup() {  
  pinMode(13, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(13, HIGH);  
  delay(1000);  
  digitalWrite(13, LOW);  
  delay(1000);  
}
```

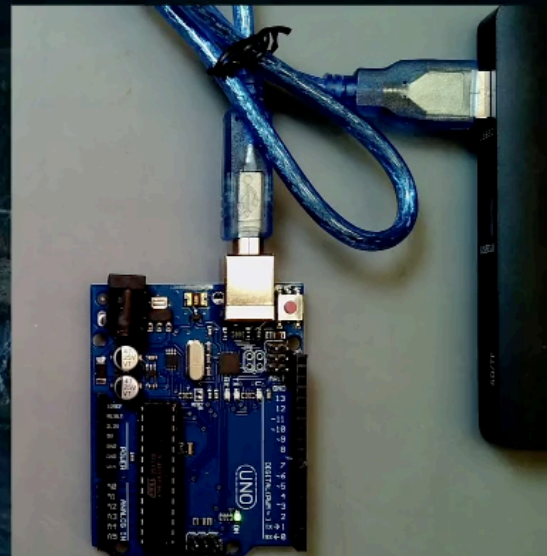
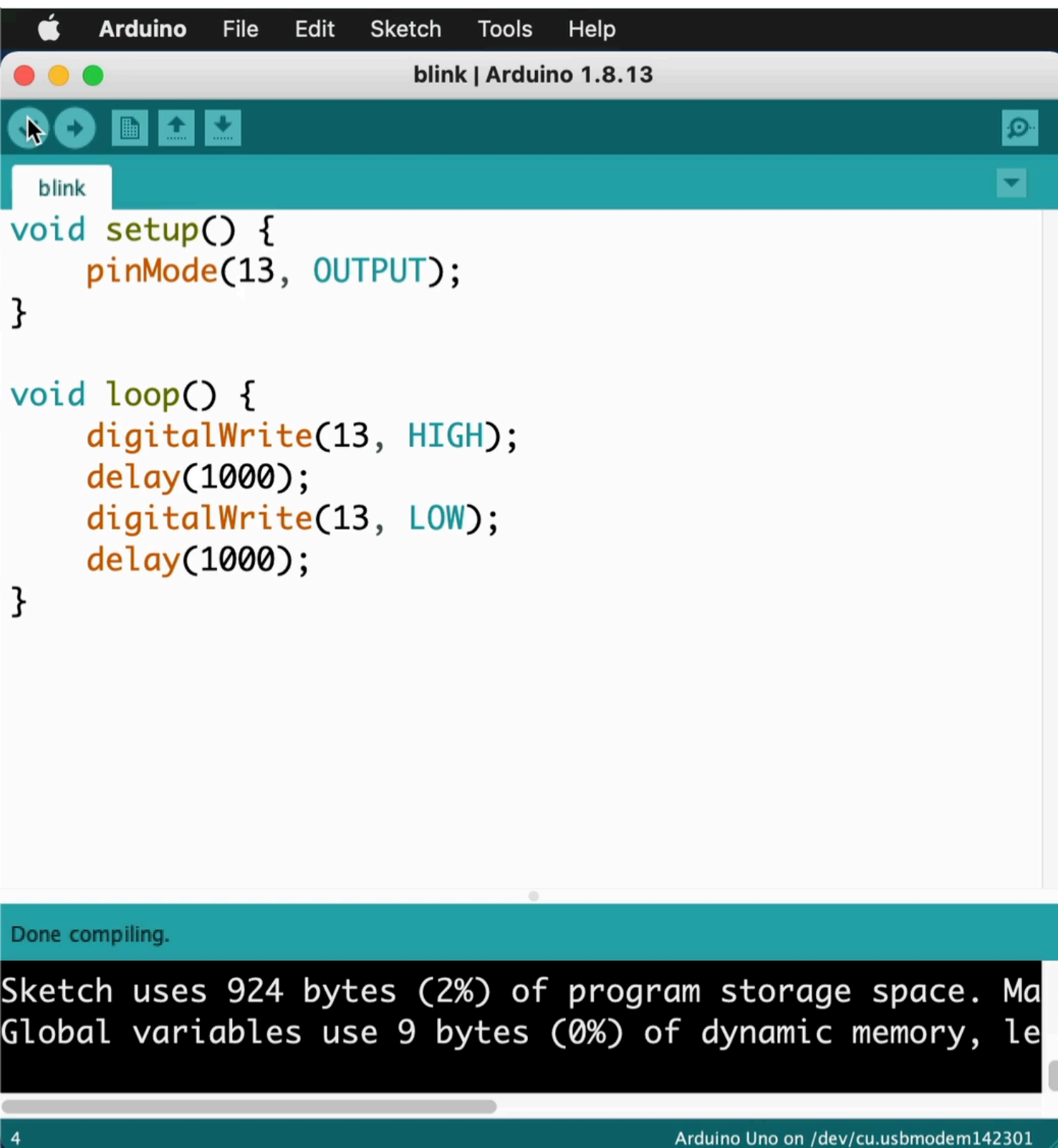
The status bar at the bottom indicates 'Arduino Uno on /dev/cu.usbmodem142301'.

Serial ports

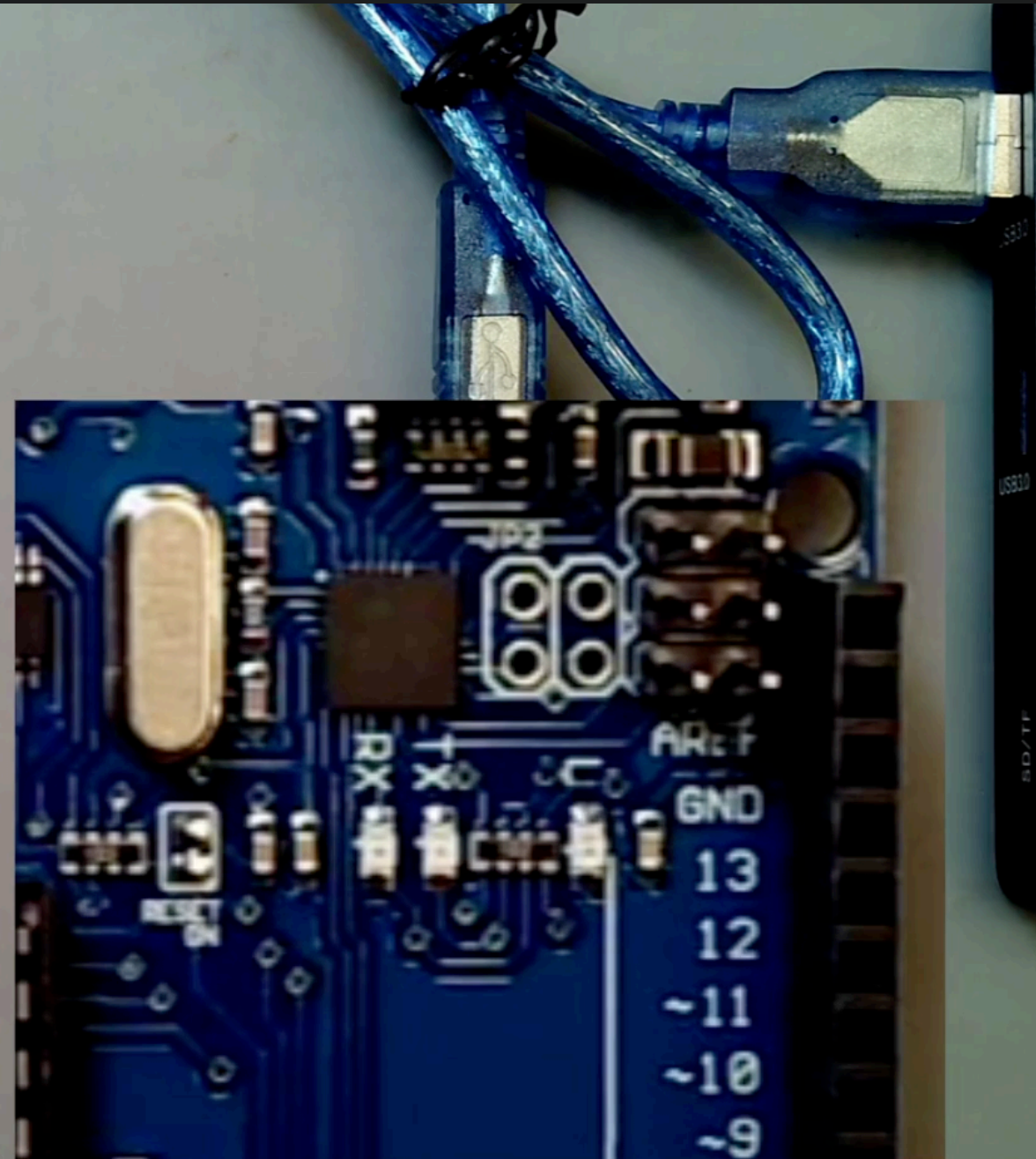
- /dev/cu.Bluetooth-Incoming-Port
- /dev/cu.JasonsBeatsStudio3-SPPS
- /dev/cu.JasonsBeatsStudio3-Wire
- ✓ /dev/cu.usbmodem142301 (Arduino Uno)

4 Arduino Uno on /dev/cu.usbmodem142301

Blink Demo



Blink Demo



Practice

Which Arduino UNO pin is connected to the onboard LED?

Respond at pollev.com/jku

A

B

C

D

A) 0

B) 5

C) 10

D) 13

Practice

What is the maximum current that can be drawn from an Arduino UNO digital output pin?

Respond at pollev.com/jku

A

B

C

D

A) 20 mA

B) 40 mA

C) 100 mA

D) 800 mA

EG1311: Design & Make

Arduino Intro

Dr. Jason S. Ku